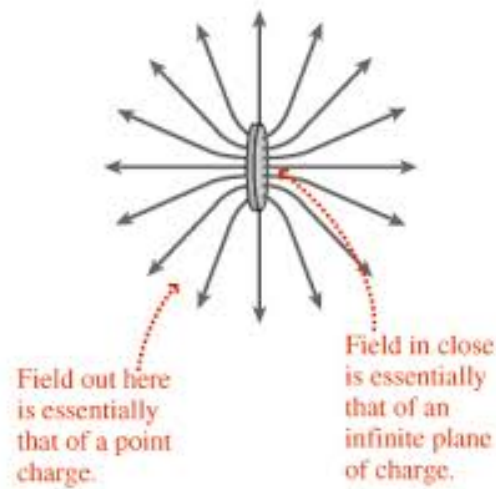


6_popquiz charged disk, dipole ...

 This is a preview of the published version of the quiz

Started: Mar 15 at 3:11pm

Quiz Instructions



Question 1 20 pts

A charged disk has a radius of $R=2.5$ cm and a surface charge density of $5.3\mu\text{C}/\text{m}^2$. See slide 95 for equation

What is the electric field on the surface of the disk? (hint: so $x = 0$)



about $3 \cdot 10^5$ N/C



$4 \cdot 10^4$ N/C



0



about 0.3 N/C



Question 2 20 pts

electric breakdown occurs when the electric field is larger than 3 Mega N/C. In that case, there is a spark.

Refer to the previous question. Will the disk spark? (Mega means 1000,000).

That is. is the electric field > 3 Mega N/C

☐

yes

☐

Whaaaaaa?

☐

It depends if the disk is positively or negatively charged

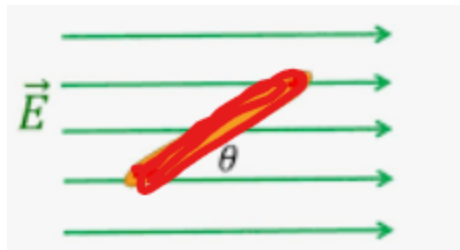
☐

no

⋮

Question 3 20 pts

seeds are placed in castor oil so we can visualize the electric field. As discussed in class. the seeds get polarized and align with the field. As discussed. Here is a seed before we turn on E (uniform)



what is true

☐

the seed does not move

☐

the seed rotates counterclockwise

☐

the seed rotates clockwise

☐

the seed accelerates along the lines

⋮

Question 4 20 pts

In one problem about computing the electric field of a linear continuous distribution of charge you get :

$$\int_0^3 \frac{72dx}{(16+x^2)^{3/2}}$$

You don't remember your calculus class but you have wolfram alpha for example. Use it to compute this integral.

(integral of for x between 0 and 3).

what do you get? replace 3/2 by 1.5

☐

2.7

☐

1

☐

1.8

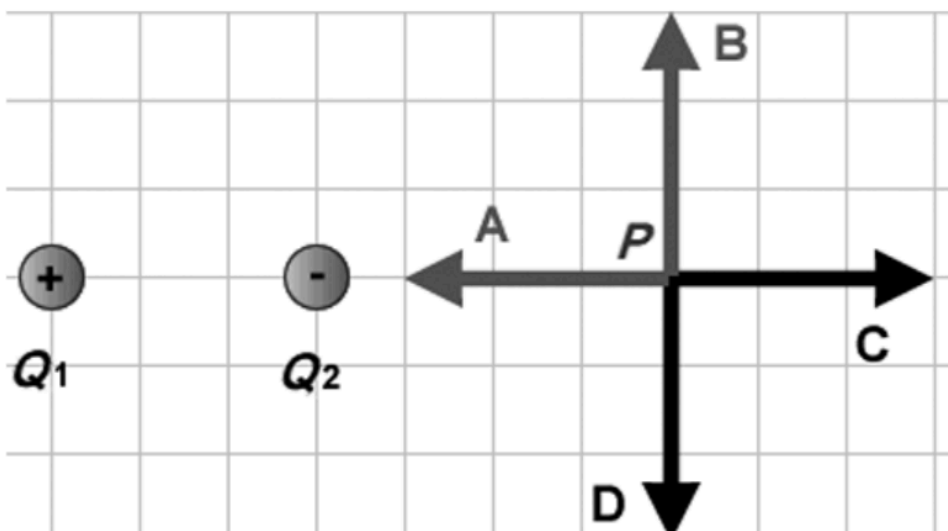
☐

5.4

⋮

Question 5 20 pts

Two point charges Q_1 and Q_2 of equal magnitudes and opposite signs are positioned as shown in the figure. Which of the arrows best represents the net electric field at point P due to these two charges?



this was a question in last year test1. you can use simulation



A



B



C



D



Question 6 5 pts

1. A long thin rod of length 50 cm has a total charge of 5 mC uniformly distributed over it. Find the linear charge density.

Hint: convert mC to C and cm to m

1 C = 1000mC and 1m = 100cm

hint: linear density is charge divided by length



0.01C/m



10C/m



0.05C/m



1C/m

Not saved

Submit Quiz